

ENGINEERING INTERN CASE STUDY

Build a Simple Task API

You have been asked to build the backend for a simple task management application. Users should be able to create tasks, view them, update them and mark them as completed.

The application does **not** need a frontend.

Your Task

Build a REST API that allows a user to:

- Create a task
- View all tasks
- View a single task
- Update a task
- Mark a task as completed
- Delete a task

Each task should contain at least:

- `id`
- `title`
- `description`
- `status`
- `createdAt`
- `updatedAt`

You may use any backend language or framework you are comfortable with.

But there's a catch...

Your API should also handle these situations properly:

A user tries to create a task without a title.

A user requests a task that does not exist.

A user tries to update a task with invalid data.

The same task should not somehow be created twice because a request was accidentally submitted twice.

Database

You may use any database you are comfortable with, including: **PostgreSQL**, **MySQL**, **MongoDB**, **SQLite** or another reasonable option. Please do not use an in-memory array as your final database.

What to Submit

Submit a link to a **GitHub repository** containing your project.

Your repository should include a README explaining:

- how to run the application;
- technologies used;
- API endpoints;
- database setup;
- assumptions you made;
- any known limitations;
- one thing you would improve if you had more time.

Also include either:

- a Postman collection;
- Swagger/OpenAPI documentation; or
- clear request/response examples in your README.

One Extra Requirement

Choose **one** of the following and implement it:

Option A: Authentication:

Users can register/login and should only be able to see their own tasks.

Option B: Filtering:

Allow tasks to be filtered by status and creation date.

Option C: Activity Log:

Keep a simple record of important actions such as task creation, completion and deletion.

Candidates should clearly state which option they selected.

Note: If you cannot explain the code you submitted, that will form part of the assessment.



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SUBMISSION GUIDELINES

- Submit your work as a PDF
- Ensure your work is original
- You may use AI tools for brainstorming, but the final submission should reflect your own thinking and writing style

DEADLINE

Please submit your completed case study within 48 to 72 hours of receiving it.

We look forward to reading your work. Good luck!

Builders' Tribe